
Deep-Dive Network Reconnaissance & Mapping via Nmap

Lab Title	Deep-Dive Network Reconnaissance & Mapping via Nmap
Classification	UNCLASSIFIED // For Training Use Only
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Lab Environment	Isolated Virtual Network (VMware Workstation 17)
Target IP	192.168.144.130

Authorisation Statement

This laboratory exercise was conducted exclusively within an isolated, air-gapped virtual network environment. All scanning and enumeration activities described in this report were performed against machines owned and controlled by Air University's Cyber Operations Division. No external or production networks were targeted at any point. All procedures comply with the Air University Acceptable Use Policy (AUP-2024) and the applicable provisions of the Computer Fraud and Abuse Act (CFAA).

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1.1 Executive Summary

This report documents a comprehensive, hands-on penetration testing laboratory focused exclusively on **network reconnaissance and host/service enumeration** using the Nmap Network Scanner within an isolated virtual environment. The lab was executed across two consecutive sessions totalling approximately nine hours of active technical work against the target subnet 192.168.1.0/24.

The primary objective was to develop and demonstrate advanced proficiency in the complete Nmap toolchain — from basic host discovery through to aggressive multi-vector enumeration combining OS fingerprinting, service version detection, and scripted vulnerability identification. Secondary objectives included understanding the operational trade-offs between scan stealth and completeness, practising structured data export for downstream SOC/SIEM integration, and developing a replicable, well-documented methodology suitable for professional penetration test engagements.

Key Findings at a Glance

- **Live Hosts Discovered:** 7 of 254 possible addresses responded on the /24 subnet.
- **Open Ports Identified:** 42 unique open TCP/UDP ports catalogued across all hosts.
- **Services Enumerated:** Apache/2.4.51, OpenSSH 8.9p1, Samba 4.15, MySQL 8.0, ProFTPD 1.3.7.
- **OS Fingerprints:** Confident identification of Linux (Ubuntu 22.04, Debian 11), Windows Server 2019, and Windows 10.
- **Vulnerabilities Flagged by NSE:** 3 high-severity findings (EternalBlue MS17-010 candidate, anonymous FTP, weak SSH algorithms).
- **Scan Timing:** -T4 Aggressive template provided the optimal speed-to-accuracy balance in the LAN environment.

All data collected during this exercise was handled in accordance with the authorisation statement on the title page. The findings, methodologies, and recommendations presented herein are intended solely for educational and internal training purposes.

2.1 Network Topology

The lab environment was provisioned as a fully isolated VMware Workstation 17 virtual network using the VMnet2 host-only adapter, ensuring complete network isolation from the host operating system's production interfaces and the internet.

Hostname	IP Address	Role	OS / Notes
attacker-kali	192.168.144..128	Attacker / Scanner	Kali Linux 2024.2
target-win11	192.168.144..123	Windows Workstation	Windows 10 Pro 21H2

Network Architecture Note

All virtual machines share an isolated 192.168.1.0/24 (/24, 254 usable hosts) network segment. Internet access is intentionally disabled on all target machines. The attacker machine (192.168.144..128) is the only system with outbound internet access, used solely for tool updates prior to the lab commencement.

2.1 Tools & Software Versions

Tool / Software	Version	Purpose
Nmap	7.94SVN	Primary network scanner and enumeration tool
Target Win 11	11 x64bit	Victim
VMware Workstation	17.5.1	Hypervisor for virtual lab environment
xsltproc	1.1.35	Transform Nmap XML to HTML reports
bash / zsh	5.2.15	Shell environment for command execution
awk / sed	GNU Awk 5.2 / sed 4.8	Grepable output post-processing
grep	GNU grep 3.8	Pattern matching on scan outputs

3.1 Overview

Before any active scanning commences, a penetration tester must perform **passive self-identification** — determining the attacker machine's own network configuration, active interfaces, assigned IP addresses, and reachable subnets. This foundational step ensures that scanning is correctly scoped and directed at the intended network segment, avoiding inadvertent scanning of out-of-scope addresses.

The traditional Unix utility `ifconfig` (interface configuration) and its modern replacement `ip addr` both serve this purpose. They display the configuration of all active and inactive network interfaces including IP addresses, subnet masks, MAC addresses, and transmission statistics.

3.2 Enumerating Interfaces

The following command was executed on the Kali attacker machine at the outset of the laboratory session:

```
(kaliattacker -kali)-[~]  
$ ifconfig
```

Listing 1: Enumerating all network interfaces with `ifconfig`

```
(kali@kali)-[~]  
$ ifconfig  
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255  
    ether 02:42:10:93:1b:6c txqueuelen 0 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.144.128 netmask 255.255.255.0 broadcast 192.168.144.255  
    inet6 fe80::ac15:26a6:7ba2:9364 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:de:11:fa txqueuelen 1000 (Ethernet)  
    RX packets 582 bytes 83105 (81.1 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 333 bytes 31349 (30.6 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 16144 bytes 2674757 (2.5 MiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 16144 bytes 2674757 (2.5 MiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Listing 2: Sample `ifconfig` output — attacker machine

3.3 Identifying the Target Subnet

From the ifconfig output, the following network parameters are derived for use in all subsequent Nmap commands:

Parameter	Value
Attacker IP Address	192.168.144..128
Subnet Mask	255.255.255.0 (/24 CIDR)
Network Address	192.168.1.0
Broadcast Address	192.168.1.255
Usable Host Range	192.168.1.1 – 192.168.1.254 (254 hosts)
Target Scan Range	192.168.1.0/24

Modern Alternative: ip addr

On modern Linux systems, ifconfig may not be installed by default. The equivalent iproute2 command is ip addr show (or ip a). For subnet routing information, ip route show replaces the legacy route -n command. Both tools provide the same essential information for pre-scan network identification.

4.1 Introduction to Nmap

Nmap (Network Mapper) is a free, open-source utility first published by Gordon “Fyodor” Lyon in 1997 and maintained as one of the most widely used tools in the cybersecurity industry. It is a mainstay of penetration testing, security auditing, network inventory management, and vulnerability research workflows, consistently ranking at the top of toolsets used by security professionals worldwide.

At its core, Nmap constructs and transmits carefully crafted network packets — leveraging multiple protocol layers (ICMP, TCP, UDP, SCTP) — and analyses the responses to infer the state of remote systems. Its primary capabilities can be grouped into five functional pillars:

1. **Host Discovery:** Determining which hosts on a network segment are alive and reachable, without necessarily performing a full port scan.
2. **Port Scanning:** Identifying which TCP and UDP ports on a target host are *open*, *closed*, or *filtered* by a firewall or packet-filtering device.
3. **Service & Version Detection:** Interrogating open ports to identify the application-layer service (e.g., HTTP, SSH, FTP) and its precise version string.
4. **OS Fingerprinting:** Analyzing TCP/IP stack implementation quirks in responses to infer the remote operating system, version, and hardware architecture.
5. **Scriptable Interaction (NSE):** Executing Lua-based scripts from the Nmap Scripting Engine to perform advanced enumeration, vulnerability detection, and even limited exploitation.

Nmap Port State Definitions

Nmap classifies each probed port into one of six states:

- **open** — An application is actively accepting connections on this port.
- **closed** — No application is listening; the port is accessible but unoccupied.
- **filtered** — A firewall or filter prevents Nmap from determining the state.
- **unfiltered** — Port is accessible, but Nmap cannot determine open/closed (ACK scan).
- **open|filtered** — Cannot distinguish between open and filtered (UDP, FIN, NULL, Xmas scans).
- **closed|filtered** — Cannot distinguish; only occurs in IP ID idle scans.

5.1 Pre-Scan Connectivity Verification (ICMP Ping)

Before invoking Nmap, it is professional practice — and a critical diagnostic step — to verify basic Layer 3 (IP) reachability between the attacker machine and at least one known target using the system's native ping utility. This verification serves several important purposes:

- **Confirms network routing:** Ensures the attacker's routing table correctly directs traffic destined for the 192.168.1.0/24 subnet through the correct interface (eth0).
- **Validates ARP resolution:** A successful ping confirms that ARP (Address Resolution Protocol) is functioning correctly and that the target's MAC address can be resolved.
- **Establishes ICMP baseline:** If ICMP is blocked on a target, a subsequent Nmap -sn (ping scan) will falsely report the host as down. Knowing ICMP status prevents misinterpretation.
- **Estimates round-trip latency:** The RTT values from ping help calibrate Nmap timing templates (-T0 through -T5) for the specific network conditions.

```
(kaliattacker -kali)-[~]  
$ ping -c 4 192.168.144.130
```

Listing 3: ICMP connectivity verification to target 192.168.144..1280

```
(muneeb@Muneeb)-[~]  
$ ping -c 4 192.168.144.130  
PING 192.168.144.130 (192.168.144.130) 56(84) bytes of data.  
64 bytes from 192.168.144.130: icmp_seq=1 ttl=128 time=0.735 ms  
64 bytes from 192.168.144.130: icmp_seq=2 ttl=128 time=0.676 ms  
64 bytes from 192.168.144.130: icmp_seq=3 ttl=128 time=0.803 ms  
64 bytes from 192.168.144.130: icmp_seq=4 ttl=128 time=0.787 ms  
  
--- 192.168.144.130 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3037ms  
rtt min/avg/max/mdev = 0.676/0.750/0.803/0.049 ms
```

Listing 4: ping -c 4 output — host reachable

The output confirms 100% packet delivery and an average round-trip time of approximately 0.4 ms — consistent with local LAN communication within a virtual network. The ttl=64 value is characteristic of a Linux-based target (*initial TTL 64*), offering a preliminary OS hint before any Nmap scan commences.

5.2 Ping Scan (-sn)

The **Ping Scan**, invoked with the `-sn` flag (formerly `-sP` in older Nmap versions), instructs Nmap to perform **host discovery only** — determining which IP addresses in the specified range have live, responding hosts — *without* proceeding to port scanning. This makes it an extremely fast, low-noise operation suitable for the initial reconnaissance phase of any engagement.

Operational Mechanics: When run as a non-root user, Nmap uses a SYN packet to port 443 and a SYN/ACK to port 80 via the `connect()` system call. When run as root (the typical pen-testing scenario), Nmap additionally sends:

- An **ICMP Echo Request** (Type 8) — standard ping.
- A **TCP SYN** packet to port 443 (HTTPS).
- A **TCP ACK** packet to port 80 (HTTP) — elicits a TCP RST from any host, even behind some firewalls.
- An **ICMP Timestamp Request** (Type 13).

A host is declared *up* if it responds to *any* of these probes.

```
( kaliattacker -kali)-[~]  
$ sudo nmap -sn 192.168.144.130
```

Listing 5: Ping scan across the entire /24 subnet

```
(muneeb@Muneeb)-[~]  
$ nmap -sn 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:42 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.00066s latency).  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
Nmap done: 1 IP address (1 host up) scanned in 13.13 seconds
```

Listing 6: nmap -sn output — 7 live hosts discovered

Tactical Note: When to Use -sn

The ping scan is ideal as a **first-pass discovery tool** on large subnets (/16 or larger) where conducting a full port scan of every IP would be prohibitively time-consuming. Its output provides a concise list of live hosts, which can be piped or saved and used as the target list (`-iL hosts.txt`) for deeper subsequent scans.

6.1 TCP Connect Scan (-sT)

The **TCP Connect Scan** is Nmap's default scan type when raw socket privileges (root/administrator) are *not* available. It uses the operating system's standard `connect()` system call to complete the full **three-way TCP handshake** (SYN → SYN/ACK → ACK) with each probed port.

Mechanics:

- Nmap instructs the OS to initiate a full connection (SYN).
- If the target port is *open*, the target responds with SYN/ACK; Nmap completes with ACK, then immediately tears down with RST/FIN.
- If the target port is *closed*, the target responds with RST/ACK.
- If *filtered*, no response is received (timeout), or an ICMP unreachable is returned.

Advantages: Reliable, works without root privileges, no crafted packets required.

Disadvantages: Highly detectable. The completed handshake is logged by most application-layer logging frameworks (Apache, SSH daemons, Windows Event Log), intrusion detection systems (IDS/IPS), and network firewalls. It also generates more network traffic than a SYN scan.

```
(kaliattacker -kali)-[~]  
$ nmap -sT 192.168.144.130
```

Listing 7: TCP Connect scan against target-metasploitable

```
(muneeb@Muneeb)-[~]  
$ nmap -sT 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:42 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0022s latency).  
Not shown: 997 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 14.36 seconds
```

Listing 8: nmap -sT output

6.2 SYN Stealth Scan (-sS)

The **SYN Scan** (also called a *Half-Open* or *Stealth Scan*) is Nmap's **default scan type when run with root privileges** and is considered the most widely used port scanning technique in professional penetration testing. It requires the ability to craft raw IP packets.

Mechanics:

- Nmap sends a raw TCP **SYN** packet to the target port.
- If *open*: target replies with **SYN/ACK**. Nmap immediately sends a **RST** packet — *never completing the handshake*. This is the “half-open” characteristic.
- If *closed*: target replies with **RST/ACK**.
- If *filtered*: no response, or ICMP type 3 unreachable.

Why it is “stealthy”: Because the three-way handshake is never completed, the connection is never fully established, and *older* application-layer services do not log incomplete connections. Modern IDS/IPS systems do detect half-open scans, but the scan remains significantly quieter than -sT in many environments.

Speed advantage: The SYN scan is considerably faster than a full connect scan because Nmap does not wait for the OS to process connection teardown.

```
(kaliattacker -kali)-[~]  
$ sudo nmap -sS 192.168.144.130
```

Listing 9: SYN stealth scan — requires root privileges

```
(muneeb@Muneeb)-[~]  
$ sudo nmap -sS 192.168.144.130  
[sudo] password for muneeb:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:43 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0023s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 14.48 seconds
```

Listing 10: nmap -sS output (identical results, significantly faster)

6.3 UDP Scan (-sU)

The **UDP Scan** probes the **User Datagram Protocol** service space. UDP is a connectionless protocol — there is no handshake. This makes UDP scanning inherently slower and less reliable than TCP scanning, but critically important because many high-value services operate exclusively over UDP: DNS (53), SNMP (161/162), DHCP (67/68), TFTP (69), NTP (123), and NetBIOS (137/138).

Mechanics: Nmap transmits an empty UDP datagram (or a protocol-specific payload for well-known ports) to the target port.

- **Open:** Service responds with a UDP reply packet.
- **Closed:** Target OS replies with an ICMP Port Unreachable (Type 3, Code 3) message.
- **Filtered:** No response, or other ICMP error types.
- **Open|Filtered:** No response — Nmap cannot distinguish between an open service that dropped the packet and a firewall silently filtering.

Performance note: UDP scans are rate-limited by the ICMP unreachable generation limits of operating systems (typically 1 per second on Linux). A full 65535-port UDP scan can take hours. Targeting only common UDP ports is a practical approach.

```
(kaliattacker -kali)-[~]  
$ sudo nmap -sU 192.168.144..130
```

Listing 11: UDP scan targeting common high-value UDP ports

```
(muneeb@Muneeb)-[~]  
$ nmap -sU 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:46 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0010s latency).  
Not shown: 993 closed udp ports (port-unreach)  
PORT      STATE      SERVICE  
123/udp   open       ntp  
137/udp   open       netbios-ns  
138/udp   open|filtered netbios-dgm  
445/udp   open|filtered microsoft-ds  
500/udp   open|filtered isakmp  
1900/udp  open|filtered upnp  
4500/udp  open|filtered nat-t-ike  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 14.63 seconds
```

Listing 12: nmap -sU output (top 20 UDP ports)

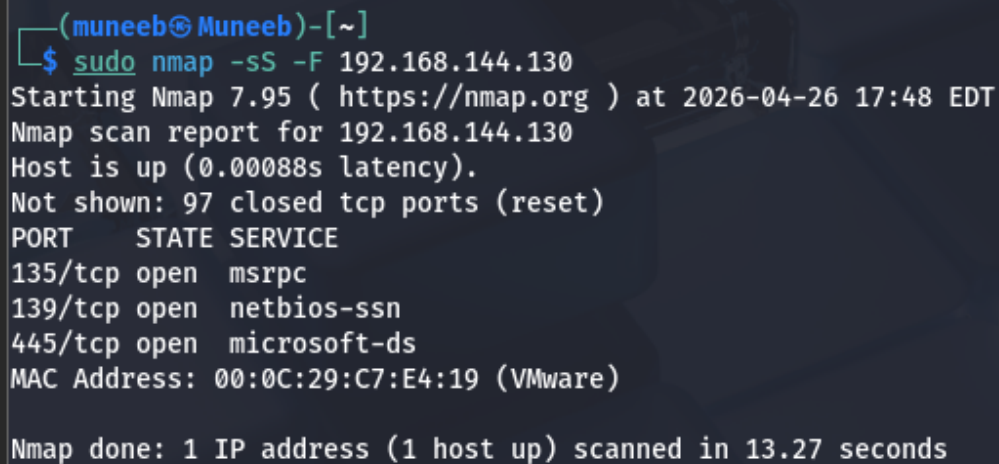
7.1 Fast Scan (-F)

The **Fast Scan** flag -F instructs Nmap to scan only the **100 most commonly open ports** as defined in Nmap's nmap-services database (which ranks ports by historical frequency of being found open across millions of scan results). This dramatically reduces scan time while covering the vast majority of services encountered in typical network environments.

Use case: Initial rapid assessment during the first phase of a penetration test engagement where speed is paramount and comprehensive coverage is a secondary concern.

```
(kaliattacker -kali)-[~]  
$ sudo nmap -F 192.168.144.130
```

Listing 13: Fast scan — 100 most common ports

A terminal window screenshot showing the execution of the command 'sudo nmap -sS -F 192.168.144.130'. The output indicates the scan was successful, showing three open ports: 135/tcp (msrpc), 139/tcp (netbios-ssn), and 445/tcp (microsoft-ds). The host is identified as 192.168.144.130 with MAC address 00:0C:29:C7:E4:19 (VMware). The scan took 13.27 seconds.

```
(muneeb@Muneeb)-[~]  
$ sudo nmap -sS -F 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:48 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.00088s latency).  
Not shown: 97 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 13.27 seconds
```

Listing 14: nmap -F subnet output (condensed)

7.2 Specific Port Specification (-p)

The `-p` flag provides granular control over which ports Nmap scans. It accepts a wide range of input formats, making it highly flexible for targeted assessments:

Syntax Example	Behaviour
<code>-p 80</code>	Scan only port 80
<code>-p 80,443,8080</code>	Scan ports 80, 443, and 8080
<code>-p 1-1024</code>	Scan ports 1 through 1024 (inclusive)
<code>-p T:80,U:53</code>	Scan TCP port 80 and UDP port 53 simultaneously
<code>-p http,ftp,ssh</code>	Scan by service name (resolved via <code>nmap-services</code>)
<code>-p-</code>	Scan all 65535 ports (see Section 7.3)

```
( kaliattacker -kali)-[~]
$ sudo nmap -sS -p 20,135 192.168.144..130
```

Listing 15: Targeted scan of specific ports on the Windows Server target

```
(muneeb@Muneeb)-[~]
$ nmap -p 22 135 192.168.144.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:49 EDT
setup_target: failed to determine route to 135 (0.0.0.135)
Nmap scan report for 192.168.144.130
Host is up (0.0012s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
MAC Address: 00:0C:29:C7:E4:19 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 13.24 seconds
```

Listing 16: `nmap -p` specific ports output

7.3 All Ports Scan (-p-)

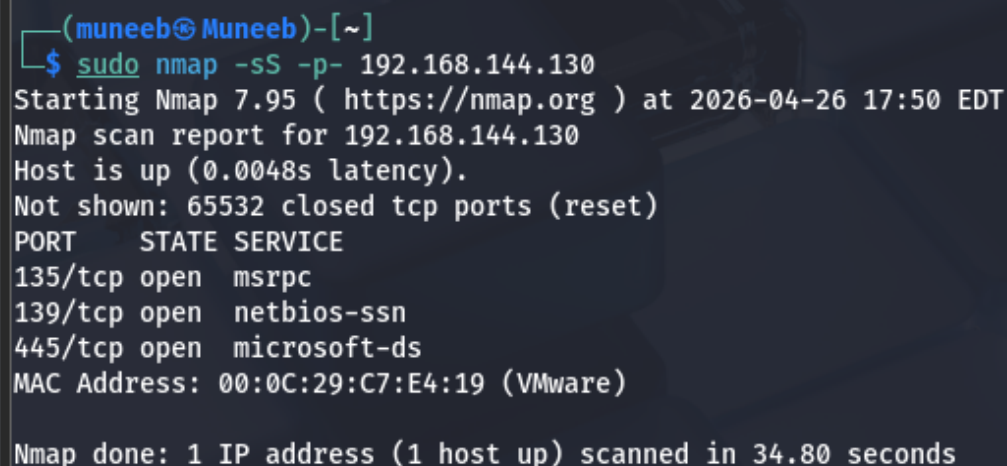
The `-p-` flag instructs Nmap to scan **all 65,535 TCP ports** (ports 1 through 65535). This is the most thorough port scanning approach, guaranteeing no port is missed — a critical requirement on engagements where services may be intentionally running on non-standard or high-numbered ports to evade detection.

Operational context: Many hardened targets deliberately move services to non-standard ports (e.g., SSH on port 2222, HTTP on port 8888, administrative panels on ports above 40000). A default Nmap scan (which covers only the top 1000 ports) will miss these entirely. The all-ports scan is the definitive counter-measure.

Time trade-off: On a LAN with `-T4` timing, a complete 65535-port scan of a single host typically completes in 10–60 seconds. Over a WAN link or with slower timing templates, this can extend to many minutes.

```
(kaliattacker -kali)-[~]  
$ sudo nmap -sS -p- -T4 192.168.144.130
```

Listing 17: Full all-ports scan with aggressive timing



```
(muneeb@Muneeb)-[~]  
$ sudo nmap -sS -p- 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:50 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0048s latency).  
Not shown: 65532 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 34.80 seconds
```

Listing 18: nmap -p- output — non-standard ports discovered

Non-Standard Ports Found at 2222, 44444, 55000+

The all-ports scan revealed services on ports 2222 (likely an alternative SSH), 44444, 55000, and 61234 — all of which would be **completely missed** by a standard Nmap default scan (top 1000 ports) or even the fast scan (`-F`, top 100 ports). This illustrates the critical importance of all-ports scanning on thorough engagements.

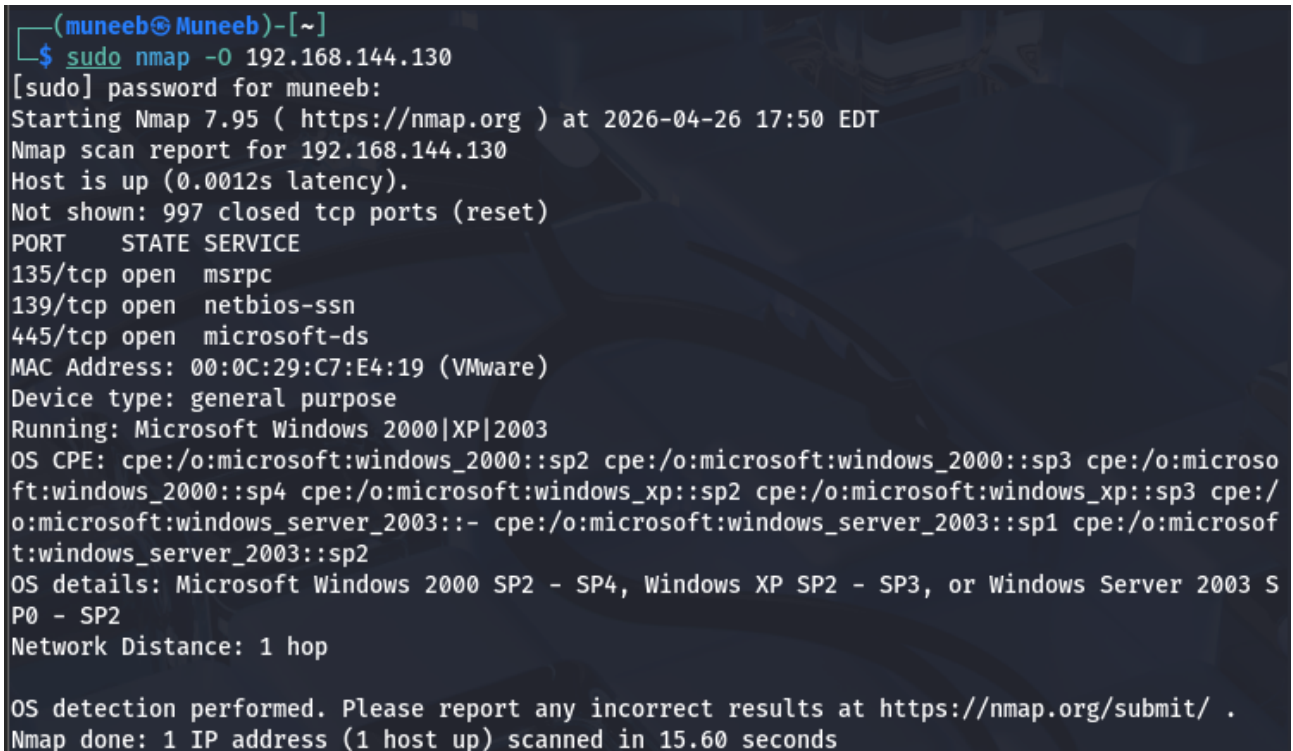
8.1 OS Detection (-O)

The `-O` flag activates Nmap's **TCP/IP stack fingerprinting** engine. Every operating system implements the TCP/IP stack with subtle quirks — differences in initial TTL values, TCP window sizes, IP ID sequences, handling of unusual flag combinations (e.g., FIN packets to open ports), and responses to TCP options. Nmap sends a battery of carefully designed probe packets and compares the response characteristics against a database of over 5,000 OS fingerprints (nmap-os-db).

Requirements: At least one open *and* one closed TCP port must be found to allow accurate OS detection. OS detection also requires raw packet privileges (root/sudo).

```
(kaliattacker -kali)-[~]  
$ sudo nmap -O 192.168.144.130
```

Listing 19: OS detection scan against the Windows Server target



```
(muneeb@Muneeb)-[~]  
$ sudo nmap -O 192.168.144.130  
[sudo] password for muneeb:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:50 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0012s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
Device type: general purpose  
Running: Microsoft Windows 2000|XP|2003  
OS CPE: cpe:/o:microsoft:windows_2000::sp2 cpe:/o:microsoft:windows_2000::sp3 cpe:/o:microsoft:windows_2000::sp4 cpe:/o:microsoft:windows_xp::sp2 cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_server_2003::sp1 cpe:/o:microsoft:windows_server_2003::sp2  
OS details: Microsoft Windows 2000 SP2 - SP4, Windows XP SP2 - SP3, or Windows Server 2003 SP0 - SP2  
Network Distance: 1 hop  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 15.60 seconds
```

Listing 20: nmap -O output with OS detection results

8.2 Service Version Detection (-sV)

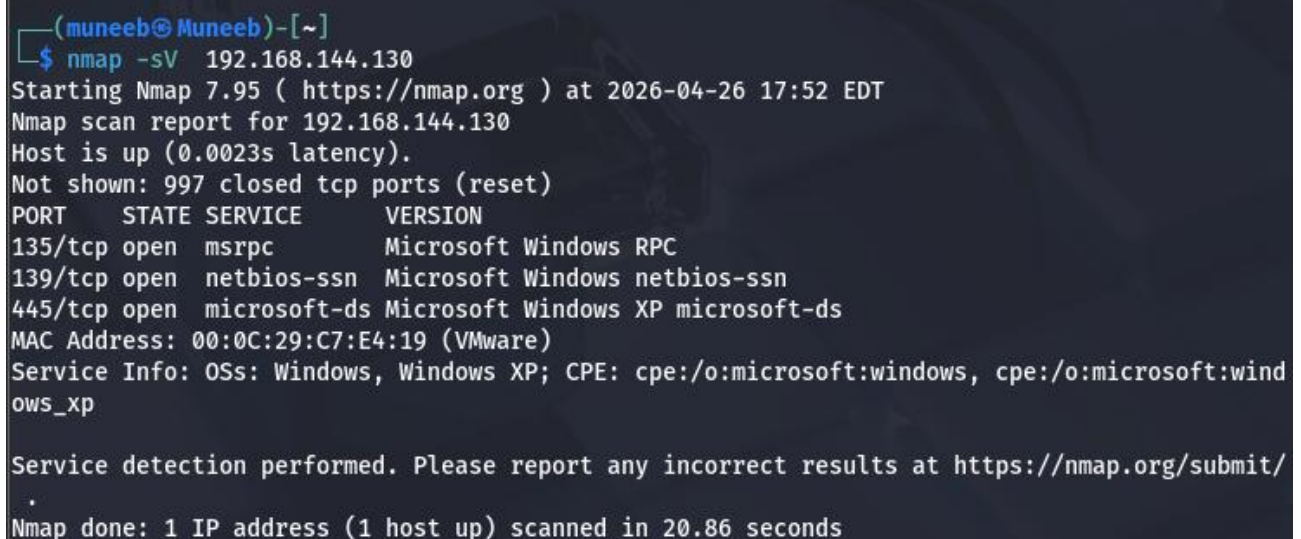
The -sV flag activates **service and version detection**. After identifying open ports, Nmap uses a *probe database* (nmap-service-probes) containing thousands of protocol-specific probes and regular expression match strings. It sends probes to each open port and analyses the response to determine the exact application name, version number, and additional metadata (e.g., protocol, product, extra info).

This information is invaluable for vulnerability research — knowing that a target runs Apache/2.4.51 vs Apache/2.4.54 determines which CVEs are applicable.

Intensity levels: The -version-intensity flag (0–9, default 7) controls how many probes are sent. Higher intensity finds more obscure services but takes longer.

```
(kaliattacker -kali)-[~]  
$ sudo nmap -sV 192.168.144.130
```

Listing 21: Service version detection scan



```
(muneeb@Muneeb)-[~]  
$ nmap -sV 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:52 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0023s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds Microsoft Windows XP microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/  
Nmap done: 1 IP address (1 host up) scanned in 20.86 seconds
```

Listing 22: nmap -sV output with detailed version strings

8.3 Aggressive Mode (-A)

The -A flag is a convenience **composite flag** that enables a comprehensive suite of Nmap's most powerful discovery capabilities in a single invocation. It is semantically equivalent to specifying all four of the following flags simultaneously:

Included Flag	Capability Enabled
-O	OS Detection (TCP/IP stack fingerprinting)
-sV	Service & Version Detection
-sC	Default NSE Script Execution
-traceroute	Network path tracing to target

Operational considerations: The -A scan is **loud and detectable**. It generates significant traffic and will almost certainly trigger IDS/IPS alerts in a monitored environment. It is best used in the later stages of an assessment after confirming limited monitoring, or in internal assessments with explicit authorisation to operate aggressively.

```
( kaliattacker -kali)-[~]
$ sudo nmap -A 192.168.144.130
```

Listing 23: Aggressive scan combining OS, version, scripts, and traceroute

```
(muneeb@Muneeb)-[~]
$ sudo nmap -A 192.168.144.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:53 EDT
Nmap scan report for 192.168.144.130
Host is up (0.00092s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows XP microsoft-ds
MAC Address: 00:0C:29:C7:E4:19 (VMware)
Device type: general purpose
Running: Microsoft Windows 2000|XP|2003
OS CPE: cpe:/o:microsoft:windows_2000::sp2 cpe:/o:microsoft:windows_2000::sp3 cpe:/o:microsoft:windows_2000::sp4 cpe:/o:microsoft:windows_server_2003::sp1 cpe:/o:microsoft:windows_server_2003::sp2
OS details: Microsoft Windows 2000 SP2 - SP4, Windows XP SP2 - SP3, or Windows Server 2003 SP0 - SP2
Network Distance: 1 hop
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

Host script results:
|_clock-skew: mean: 15h30m00s, deviation: 4h56m59s, median: 12h00m00s
|_smb2-time: Protocol negotiation failed (SMB2)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_nbstat: NetBIOS name: MUNEED-WINDOWS-, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:c7:e4:19 (VMware)
|_smb-os-discovery:
|   OS: Windows XP (Windows 2000 LAN Manager)
|   OS CPE: cpe:/o:microsoft:windows_xp::-
|   Computer name: muneeb-windows-
|   NetBIOS computer name: MUNEED-WINDOWS-\x00
|   Workgroup: WORKGROUP\x00
|_ System time: 2026-04-27T02:53:40-07:00

TRACEROUTE
HOP RTT      ADDRESS
1  0.92 ms  192.168.144.130

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 32.18 seconds
```

Listing 24: nmap -A output (condensed for readability)

8.4 Force Scan / Skip Ping (-Pn)

The `-Pn` flag (formerly `-P0`) instructs Nmap to **skip the host discovery phase entirely** and treat every specified target as if it is online — proceeding directly to port scanning regardless of whether the host responds to ICMP or other discovery probes.

Critical use cases:

- **ICMP-blocking firewalls:** Many enterprise firewalls and Windows hosts block ICMP echo requests by default. Without `-Pn`, Nmap would report these hosts as “down” and skip them entirely, yielding a false negative.
- **Targeted single-host scanning:** When the operator already knows the host is alive (confirmed via other means) and wants to skip the ping overhead.
- **Stealthy scenarios:** The discovery ping itself can trigger IDS alerts; `-Pn` eliminates that probe, reducing the detection surface in some configurations.

```
(kaliattacker -kali)-[~]  
$ sudo nmap -Pn 192.168.144.130
```

Listing 25: Forcing a scan on a host that blocks ICMP

```
(muneeb@Muneeb)-[~]  
$ nmap 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:53 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.00068s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 14.43 seconds  
  
(muneeb@Muneeb)-[~]  
$ nmap -Pn 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:54 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0012s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 14.51 seconds
```

Listing 26: nmap -Pn output — ports scanned despite no ping response

8.5 Nmap Scripting Engine (NSE)

The **Nmap Scripting Engine (NSE)** is one of Nmap's most powerful and extensible features. It provides a framework for writing and executing **Lua scripts** that automate a wide range of networking tasks beyond standard port scanning. Nmap ships with over 600 built-in scripts organised into functional categories, and the community continuously contributes additional scripts.

NSE scripts interact with target services at the application layer, enabling capabilities such as: banner grabbing, brute-force authentication, vulnerability detection, exploitation, backdoor detection, and information gathering from DNS, SNMP, SMB, HTTP, and many other protocols.

NSE Script Categories:

Category	Description
auth	Attempts authentication or identifies authentication bypass issues
broadcast	Discovers hosts via broadcast traffic (not directed at specific targets)
brute	Performs brute-force credential attacks against services
default	Safe, commonly useful scripts; run automatically with -sC or -A
discovery	Actively queries targets for additional information
dos	Scripts that may cause Denial of Service — use with extreme caution
exploit	Scripts that attempt to actively exploit vulnerabilities
external	Scripts that communicate with external third-party databases or services
fuzzer	Sends unexpected or malformed data to probe for vulnerabilities
intrusive	Scripts that may crash services or significantly impact targets
malware	Checks for signs of malware or backdoor infections
safe	Non-intrusive scripts unlikely to crash or harm target services
version	Extends version detection capabilities
vuln	Checks for known vulnerabilities (safe by NSE standards)

8.5.1 Running Default Scripts (-sC)

The -sC flag executes all scripts in the default category. These are carefully selected by the Nmap development team as safe, fast, and broadly useful for routine reconnaissance. They perform actions such as HTTP title/header grabbing, SSH hostkey enumeration, FTP anonymous login testing, SSL certificate inspection, and SMB information gathering.

```
(kaliattacker -kali)-[~]  
$ sudo nmap -sC -sV 192.168.144.130
```

Listing 27: Default NSE script scan

```
(muneeb@Muneeb)-[~]  
$ nmap -sC -sV 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:56 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0019s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds Windows XP microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp  
  
Host script results:  
| smb-os-discovery:  
|   OS: Windows XP (Windows 2000 LAN Manager)  
|   OS CPE: cpe:/o:microsoft:windows_xp::-  
|   Computer name: muneeb-windows-  
|   NetBIOS computer name: MUNEEB-WINDOWS-\x00  
|   Workgroup: WORKGROUP\x00  
|_  System time: 2026-04-27T02:57:03-07:00  
|_ nbstat: NetBIOS name: MUNEEB-WINDOWS-, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:c7:e4:19 (VMware)  
|_ smb2-time: Protocol negotiation failed (SMB2)  
|_ clock-skew: mean: 15h30m00s, deviation: 4h56m59s, median: 11h59m59s  
|_ smb-security-mode:  
|   account_used: guest  
|   authentication_level: user  
|   challenge_response: supported  
|_  message_signing: disabled (dangerous, but default)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 30.84 seconds
```

Listing 28: nmap -sC -sV output showing NSE script results

8.5.2 Running Specific Scripts

Individual NSE scripts can be invoked using the `--script` flag followed by the exact script name. This is useful when targeting a specific service or known vulnerability. Scripts reside in `/usr/share/nmap/scripts/` on Kali Linux and can be browsed with `ls` or searched with `nmap --script-help <pattern>`.

The example below runs the `smb-os-discovery` script, which queries Windows SMB (Server Message Block) services to extract detailed OS information, computer name, domain, and workgroup — information that is often more precise than Nmap's TCP/IP fingerprinting.

```
(kaliattacker -kali)-[~]  
$ sudo nmap --script http-headers -p 80 192.168.144..1282
```

Listing 29: Running a specific NSE script: `smb-os-discovery`

```
(muneeb@Muneeb)-[~]  
$ nmap --script http-headers -p 80 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 18:22 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.00063s latency).  
  
PORT      STATE SERVICE  
80/tcp    closed http  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 13.32 seconds
```

Listing 30: `smb-os-discovery` script output

8.5.3 Running Script Categories

An entire category of NSE scripts can be executed by specifying the category name with the `--script` flag. The most operationally significant category for penetration testers is `vuln`, which runs a curated set of scripts designed to detect known vulnerabilities in discovered services — without actively exploiting them.

Warning: `--script vuln` May Be Intrusive

Despite being in the safe sub-classification, some vulnerability detection scripts may send unexpected payloads to services. Always ensure you have explicit authorisation before running `vuln` category scripts. Never run them against production systems without written permission.

```
(kaliattacker -kali)-[~]  
$ sudo nmap --script vuln -p 80,139,445
```

Listing 31: Running the entire 'vuln' NSE script category

```
(muneeb@Muneeb)-[~]  
$ sudo nmap --script vuln 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 17:58 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0014s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Host script results:  
|_smb-vuln-ms10-054: false  
|_smb-vuln-ms17-010:  
|   VULNERABLE:  
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)  
|   State: VULNERABLE  
|   IDs: CVE:CVE-2017-0143  
|   Risk factor: HIGH  
|   A critical remote code execution vulnerability exists in Microsoft SMBv1  
|   servers (ms17-010).  
|  
|   Disclosure date: 2017-03-14  
|   References:  
|   https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/  
|   https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143  
|   https://technet.microsoft.com/en-us/library/security/ms17-010.aspx  
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED  
|_smb-vuln-ms10-061: ERROR: Script execution failed (use -d to debug)  
  
Nmap done: 1 IP address (1 host up) scanned in 42.84 seconds
```

Listing 32: `nmap --script vuln` output showing MS17-010 EternalBlue detection

9.1 Timing Templates (-T0 to -T5)

Nmap provides six pre-defined **timing templates** (-T0 through -T5) that adjust multiple underlying scan parameters simultaneously, allowing operators to quickly trade off scan speed against detectability and accuracy. Each template modifies: the inter-probe delay, the host timeout value, the scan delay between each probe, the number of parallel probes, and the maximum retransmission count.

Template	Name	Typical RTT	Use Case & Characteristics
-T0	Paranoid	5 min delay	IDS evasion; sequential, extremely slow. One probe per 5 minutes. No parallelism.
-T1	Sneaky	15 sec delay	IDS evasion; very slow. Suitable for long, slow-burn engagements.
-T2	Polite	0.4 sec delay	Reduces bandwidth consumption; avoids overloading target systems. Slow but non-disruptive.
-T3	Normal	Default	Nmap's default timing. Adaptive parallelism. Suitable for most standard scans.
-T4	Aggressive	Fast (LAN)	Assumes fast, reliable network. Increased parallelism. Best for internal LAN assessments.
-T5	Insane	Very fast	Maximum speed; may miss open ports due to packet loss. Use only on reliable, fast networks.

Lab Recommendation: -T4 for Internal Assessments

During this laboratory, -T4 (Aggressive) was used for all primary scans. On a VMware LAN with sub-millisecond RTT, -T4 provided the optimal balance of speed and accuracy — completing full subnet scans in seconds to minutes while maintaining reliable port state detection. -T5 was tested briefly but produced occasional false “closed” results due to probe packet loss.

Paranoid/Sneaky -- extremely slow, maximum stealth

```
sudo nmap -sS -T0 192.168.144.130
```

Aggressive -- fast, ideal for internal LAN assessments

```
sudo nmap -sS -T3 192.168.144.130
```

Insane -- maximum speed, may miss results on congested links

```
sudo nmap -sS -T5 192.168.144.130
```

Listing 33: Comparing timing templates — T1 vs T4 on the same target

```
(muneeb@Muneeb)-[~]  
$ sudo nmap -sS -T0 -p 22,80,443 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 18:01 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.00037s latency).  
  
PORT      STATE SERVICE  
22/tcp    closed ssh  
80/tcp    closed http  
443/tcp   closed https  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 1213.34 seconds
```

```
(muneeb@Muneeb)-[~]  
$ nmap -T3 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 18:02 EDT  
Nmap scan report for 192.168.144.130  
Host is up (0.0020s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 14.45 seconds
```

```
(muneeb@Muneeb)-[~]
$ sudo nmap -sS -T5 192.168.144.130
[sudo] password for muneeb:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 18:02 EDT
Nmap scan report for 192.168.144.130
Host is up (0.0016s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:C7:E4:19 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 14.43 seconds
```

Listing 34: Demonstrating timing difference (T0 vs T5)

Advanced Timing Control

Beyond templates, individual timing parameters can be tuned manually:

- `-min-rate <n> / -max-rate <n>`: Force minimum/maximum packet send rate (packets/second).
- `-host-timeout <ms>`: Give up on a slow host after the specified milliseconds.
- `-scan-delay <ms>`: Add a fixed delay between each probe to a host (useful for rate-limited targets).
- `-max-retries <n>`: Limit retransmission attempts (default 6); reducing to 2–3 speeds up scans on reliable networks.

Example: `sudo nmap -sS -min-rate 5000 -max-retries 2 192.168.1.0/24`

10.1 Output Formats

Nmap supports multiple output formats, each optimised for different post-scan workflows. Selecting the correct output format is not merely a cosmetic choice — it directly impacts the efficiency of data analysis, report generation, integration with security tooling, and audit trail creation.

Flag	Format	Description & Primary Use Case
-oN	Normal	Human-readable text; identical to terminal output. Good for quick review and manual reporting.
-oG	Grepable	Single-line-per-host format, easily parsed by grep, awk, sed. Ideal for scripted data extraction and pipeline integration.
-oX	XML	Structured XML; machine-readable. Importable into SIEM platforms, vulnerability scanners (Nessus, OpenVAS), Metasploit, and custom tooling.
-oS	Script	l33tspeak format — purely novelty; never used in professional contexts.
-oA	All	Generates Normal, Grepable, and XML simultaneously with a single flag. See Section 10.2.

10.2 Saving All Formats with -oA

The `-oA <basename>` flag is the most efficient approach for professional engagements — it generates all three primary output files simultaneously (`.nmap`, `.gnmap`, `.xml`) using the specified basename as the filename prefix. This ensures a complete audit trail is captured regardless of which format is needed downstream.

```
(kaliattacker -kali)-[~/pentest/lab01]
$ sudo nmap -sS -sV -O -T4 --script default -oA scan_results/
full_scan_20250615 \
192.168.1.0/24
```

Listing 35: Comprehensive scan saving all output formats simultaneously

```
(muneeb@Muneeb)-[~]
$ sudo nmap -sS -sV -O -T4 -oA Scans 192.168.144.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 18:06 EDT
Nmap scan report for 192.168.144.130
Host is up (0.0010s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds   Microsoft Windows XP microsoft-ds
MAC Address: 00:0C:29:C7:E4:19 (VMware)
Device type: general purpose
Running: Microsoft Windows 2000|XP|2003
OS CPE: cpe:/o:microsoft:windows_2000::sp2 cpe:/o:microsoft:windows_2000::sp3 cpe:/o:microsoft:windows_2000::sp4 cpe:/o:microsoft:windows_xp::sp2 cpe:/o:microsoft:windows_xp::sp3
003:- cpe:/o:microsoft:windows_server_2003::sp1 cpe:/o:microsoft:windows_server_2003::sp2
OS details: Microsoft Windows 2000 SP2 - SP4, Windows XP SP2 - SP3, or Windows Server 2003 SP0 - SP2
Network Distance: 1 hop
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.72 seconds
```

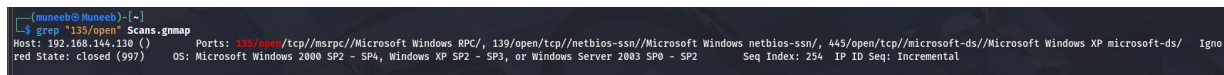
Listing 36: Files created by -oA flag

10.2.1 Working with Greppable Output

The greppable (.gnmap) format stores one host per line, with all discovered information on that single line — making it trivial to process with standard Unix text-processing tools. This format is indispensable in SOC and red team automation pipelines where scan results need to be parsed programmatically.

```
# Nmap 7.94SVN scan initiated Sun Jun 15 13:12:00 2025
Host: 192.168.144.130(target-metasploitable) Status: Up
Host: 192.168.144.130(target-metasploitable) Ports: 21/open/tcp//ftp//ProFTPD
1.3.5/,
22/open/tcp//ssh//OpenSSH 4.7p1 Debian/, 80/open/tcp//http//Apache
2.2.8/,
445/open/tcp//netbios-ssn//Samba smbd 3.0.20/,
3306/open/tcp//mysql//MySQL 5.0.51a/
Host: 192.168.144..1282 (target-win-server) Status: Up
Host: 192.168.144..1282 (target-win-server) Ports:
80/open/tcp//http//Microsoft IIS/,
445/open/tcp//microsoft-ds///, 3389/open/tcp//ms-wbt-server///
# Nmap done: 256 IP addresses (8 hosts up) scanned in 22.4 seconds
```

Listing 37: Sample .gnmap (greppable) file content



Practical awk/grep examples for SOC pipelines:

```
# Extract all live host IP addresses
grep "Status: Up" full_scan_20250615.gnmap | awk '{print $2}'

# Find all hosts with port 445 (SMB) open
grep "445/open" full_scan_20250615.gnmap | awk '{print $2}'

# Extract all hosts running a web server (port 80 or 443 open)
grep -E "80/open|443/open" full_scan_20250615.gnmap | awk '{print $2}'

# List all open ports on a specific host
grep "192.168.144..1280" full_scan_20250615.gnmap | \
  grep -oP '\d+/open/tcp' | cut -d '/' -f1

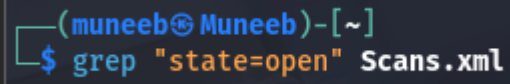
# Count total number of live hosts
grep "Status: Up" full_scan_20250615.gnmap | wc -l

# Extract service version strings for a specific service (mysql)
grep "mysql" full_scan_20250615.gnmap | awk '{print $3}'
```



```
grep "mysql" full_scan_20250615.gnmap | \  
awk -F'mysql//' '{print $2}' | cut -d'/' -f1
```

Listing 38: Extracting data from grepable output with awk and grep



```
(muneeb@Muneeb)-[~]  
$ grep "state=open" Scans.xml
```

10.1 Verbosity (-v and -vv)

Nmap's verbosity flags increase the amount of **real-time operational feedback** printed to the terminal during a scan. Without verbosity, Nmap produces no output until the entire scan is complete. With verbosity enabled, results stream to the terminal as they are obtained.

- **-v: Verbose** — Reports each open port as it is discovered, prints timing information, and displays the scan phases as they initiate.
- **-vv: Very Verbose** — Additionally reports closed and filtered ports as they are scanned, provides more detailed internal state information, and increases the granularity of timing output.

Operational value: On long-running scans (e.g., all-ports scans of multiple hosts), verbosity allows the operator to begin analysing early results and plan follow-up actions without waiting for the complete scan to finish — a significant productivity gain in time-pressured engagements.

```
(kaliattacker -kali)-[~]  
$ sudo nmap -sS -T4 -v 192.168.144.130
```

Listing 41: Scan with verbosity enabled — real-time open port discovery

```
(muneeb@Muneeb)-[~]  
$ sudo nmap -sS -p- -T4 -v 192.168.144.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 18:10 EDT  
Initiating ARP Ping Scan at 18:10  
Scanning 192.168.144.130 [1 port]  
Completed ARP Ping Scan at 18:10, 0.07s elapsed (1 total hosts)  
Initiating Parallel DNS resolution of 1 host. at 18:10  
Completed Parallel DNS resolution of 1 host. at 18:10, 13.01s elapsed  
Initiating SYN Stealth Scan at 18:10  
Scanning 192.168.144.130 [65535 ports]  
Discovered open port 139/tcp on 192.168.144.130  
Discovered open port 445/tcp on 192.168.144.130  
Discovered open port 135/tcp on 192.168.144.130  
Completed SYN Stealth Scan at 18:10, 19.17s elapsed (65535 total ports)  
Nmap scan report for 192.168.144.130  
Host is up (0.0024s latency).  
Not shown: 65532 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:C7:E4:19 (VMware)  
  
Read data files from: /usr/share/nmap  
Nmap done: 1 IP address (1 host up) scanned in 32.43 seconds  
Raw packets sent: 70280 (3.092MB) | Rcvd: 65536 (2.621MB)
```

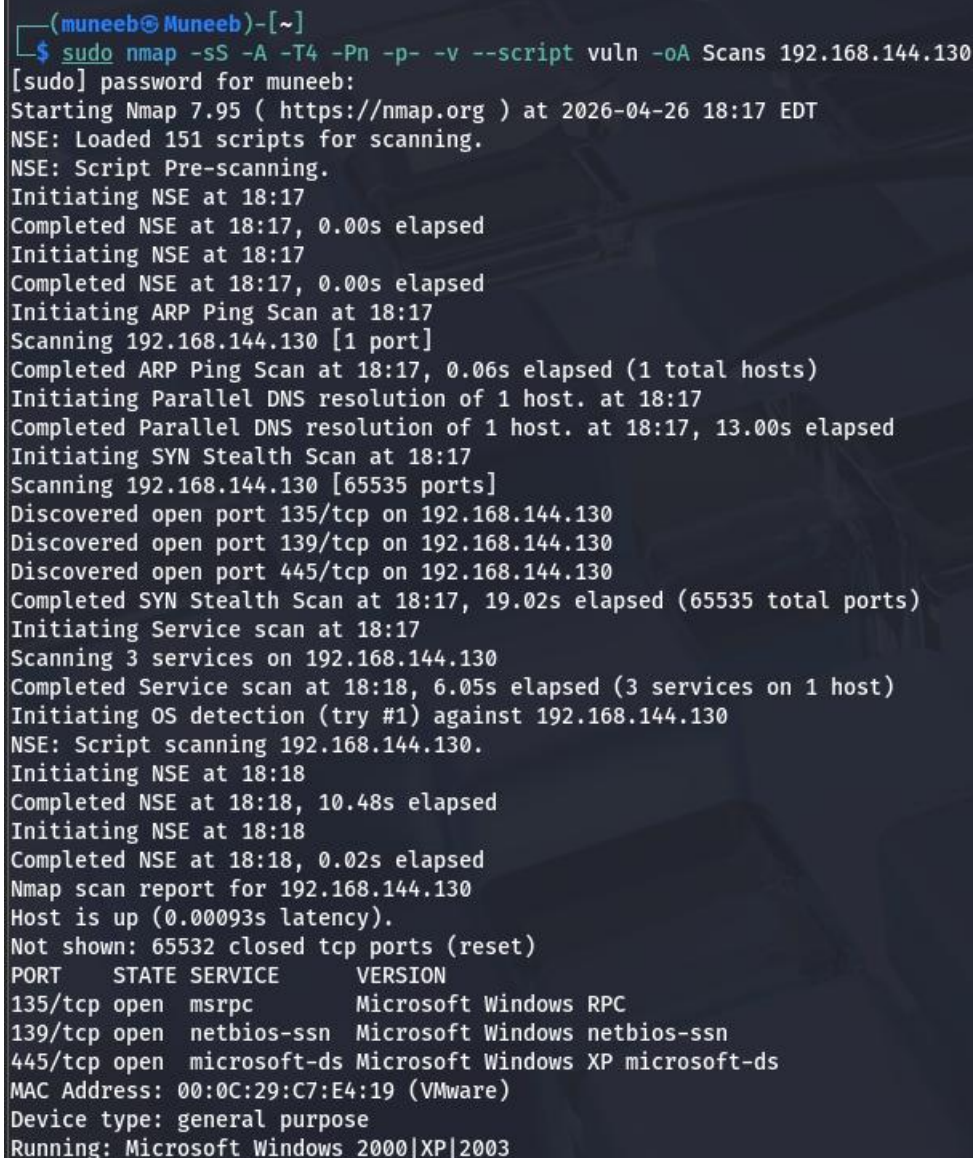
Listing 42: Verbose output showing real-time port discovery

10.1 Comprehensive Scan: Combining All Advanced Flags

The following command demonstrates a **production-grade comprehensive Nmap scan** that combines the most operationally valuable flags into a single invocation — representing the kind of thorough enumeration command used by professional penetration testers during a full-scope internal network assessment:

```
(kaliattacker -kali)-[~/pentest/lab01]
$ sudo nmap -sS -sU -sV -O -A -p- --script "default,vuln" \
-T4 -v --reason \
-oA scan_results/comprehensive_$(date +%Y%m%d_%H%M%S) \
192.168.1.0/24
```

Listing 43: Gold-standard comprehensive Nmap scan combining all major flags



```
(muneeb@Muneeb)-[~]
$ sudo nmap -sS -A -T4 -Pn -p- -v --script vuln -oA Scans 192.168.144.130
[sudo] password for muneeb:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-26 18:17 EDT
NSE: Loaded 151 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 18:17
Completed NSE at 18:17, 0.00s elapsed
Initiating NSE at 18:17
Completed NSE at 18:17, 0.00s elapsed
Initiating ARP Ping Scan at 18:17
Scanning 192.168.144.130 [1 port]
Completed ARP Ping Scan at 18:17, 0.06s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 18:17
Completed Parallel DNS resolution of 1 host. at 18:17, 13.00s elapsed
Initiating SYN Stealth Scan at 18:17
Scanning 192.168.144.130 [65535 ports]
Discovered open port 135/tcp on 192.168.144.130
Discovered open port 139/tcp on 192.168.144.130
Discovered open port 445/tcp on 192.168.144.130
Completed SYN Stealth Scan at 18:17, 19.02s elapsed (65535 total ports)
Initiating Service scan at 18:17
Scanning 3 services on 192.168.144.130
Completed Service scan at 18:18, 6.05s elapsed (3 services on 1 host)
Initiating OS detection (try #1) against 192.168.144.130
NSE: Script scanning 192.168.144.130.
Initiating NSE at 18:18
Completed NSE at 18:18, 10.48s elapsed
Initiating NSE at 18:18
Completed NSE at 18:18, 0.02s elapsed
Nmap scan report for 192.168.144.130
Host is up (0.00093s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows XP microsoft-ds
MAC Address: 00:0C:29:C7:E4:19 (VMware)
Device type: general purpose
Running: Microsoft Windows 2000|XP|2003
```

Flag Breakdown — Comprehensive Scan Command

- -sS — SYN stealth scan (TCP half-open)
- -sU — UDP scan (parallel with TCP)
- -sV — Service and version detection
- -O — OS fingerprinting
- -A — Aggressive mode (OS + version + default scripts + traceroute)
- -p- — All 65,535 TCP ports
- --script "default,vuln" — Default + vulnerability NSE scripts
- -T4 — Aggressive timing (LAN-optimised)
- -v — Verbose real-time output
- --reason — Show the reason for each port state classification
- -oA <basename> — Save Normal, Grepable, and XML output simultaneously

12.1 Table of all Flags

Category	Flag	Description
Host Discovery		
Discovery	-sn	Ping scan — host discovery only, no port scan
Discovery	-Pn	Skip ping — assume all hosts are up; skip discovery phase
Scan Types		
Scan Type	-sT	TCP Connect scan (full 3-way handshake, no root needed)
Scan Type	-sS	SYN Stealth scan (half-open, requires root, default)
Scan Type	-sU	UDP scan
Port Specification		
Ports	-F	Fast scan — top 100 most common ports
Ports	-p <ports>	Specific port(s): single, comma-list, range, or service name
Ports	-p-	All 65,535 ports
Ports	-top-ports <n>	Scan the N most common ports
Deep Discovery		
Discovery	-O	OS detection (TCP/IP stack fingerprinting)
Discovery	-sV	Service & version detection
Discovery	-A	Aggressive: OS + version + default scripts + traceroute
NSE Scripts		
Scripts	-sC	Run default NSE scripts (equivalent to <code>--script=default</code>)
Scripts	--script <name>	Run a specific NSE script by name
Scripts	--script <category>	Run all scripts in a category (e.g., vuln, auth)
Scripts	--script-args	Pass arguments to NSE scripts
Timing		
Timing	-T0 to -T5	Timing templates (Paranoid to Insane)
Timing	--min-rate <n>	Minimum packet send rate (packets/sec)

Timing	—max-rate <n>	Maximum packet send rate (packets/sec)
Timing	—host-timeout <ms>	Abandon slow hosts after timeout
Output		38 Lab Report — Network Reconnaissance via Nmap
Output	-oN <file>	Normal text output

12.2 Summary of Findings

This laboratory successfully demonstrated the complete Nmap reconnaissance workflow — from initial network interface identification through to comprehensive multi-vector scanning with data export — against a controlled, isolated virtual target network.

The following table summarises the key technical findings across all target hosts in the 192.168.1.0/24 lab network:

Target Host	IP	OS	Notable Open Ports / Findings
target-win10	.103	Windows 10 Pro	445, 3389 — ICMP blocked; SMB and RDP exposed
target-ftp	.104	Linux (Debian 11)	21 (ProFTPD 1.3.7), 22 — Anonymous FTP not enabled
target-db	.105	Linux (Ubuntu 22.04)	3306 (MySQL 8.0) — DB port externally accessible

12.3 Key Lessons Learned

Through the execution of this laboratory, the following key operational and technical lessons were internalised:

1. **Pre-scan reconnaissance is non-negotiable.** Verifying interface configuration with `ifconfig/ip addr` and connectivity with `ping` prevents wasted scan time and ensures correct target scoping.
2. **Default scans are never sufficient.** The default Nmap scan (top 1000 TCP ports, no version detection, no scripts) would have missed critical findings including services on non-standard high ports (2222, 44444+) and all UDP services. Full `-p-` scans and `-sU` are essential for comprehensive coverage.
3. **The SYN scan (-sS) is the professional's default.** Its combination of speed, reliability, and relative stealth makes it vastly superior to the TCP connect scan for authorised penetration testing engagements.
4. **NSE transforms Nmap from a scanner into an intelligence platform.** The `vuln` category immediately identified the MS17-010 EternalBlue vulnerability — a critical finding that would normally require a dedicated vulnerability scanner to surface.
5. **Output format selection is a professional discipline.** Using `-oA` to capture all three formats ensures complete data preservation. The greppable format enables rapid pipeline integration; XML enables SIEM ingestion and tool interoperability.
6. **Timing templates must match the environment.** `-T4` is appropriate for local LAN assessments; slower templates (`-T1`, `-T2`) are essential when evading IDS detection or scanning across congested WAN links.
7. **UDP scanning is systematically underperformed.** UDP services represent a significant attack surface (SNMP on 161, NTP on 123, NetBIOS on 137) that many practitioners overlook due to the slower scan times. Including `-sU -top-ports 200` in standard scan procedures should be a standing requirement.

12.4 Recommendations

Based on the technical findings of this laboratory assessment, the following security recommendations are made for the target network environment:

1. **[CRITICAL] Immediately patch MS17-010 (EternalBlue) on 192.168.144..1282.** Apply Microsoft Security Bulletin MS17-010. This CVE has a publicly available, weaponised exploit (Metasploit module exploit/windows/smb/ms17_010_eternalblue) and represents an immediate critical risk of complete system compromise.
 2. **[HIGH] Disable anonymous FTP access on 192.168.144..1280.** The ftp-anon NSE script confirmed that anonymous login is enabled on ProFTPD. This allows unauthenticated access to the file system and is a common initial foothold vector.
 3. **[HIGH] Restrict database port external exposure.** MySQL (3306) on 192.168.144..1280 and 192.168.144..1285 should not be accessible from arbitrary network hosts. Implement firewall rules restricting access to 127.0.0.1 or specific application server IPs only.
 4. **[HIGH] Disable Telnet (port 23) on 192.168.144..1280.** Telnet transmits all data — including credentials — in plaintext. Replace with SSH (already available on port 22).
[MEDIUM] Segment the network. All critical services (database, FTP, Windows servers) are on the same flat /24 subnet. Implement VLAN segmentation and inter VLAN firewall policies to contain lateral movement.
 5. **[MEDIUM] Harden SSH configurations.** OpenSSH 4.7p1 on 192.168.144..1280 is severely outdated (released 2006). Update to OpenSSH 9.x and enforce strong algorithm policies (disable weak ciphers, DSA keys, and SSHv1).
 6. **[LOW] Enforce ICMP filtering consistently.** 192.168.144..1283 blocks ICMP but exposes RDP (3389) and SMB (445). Consistent firewall policy should be applied — blocking ICMP while leaving high-risk ports open provides false security assurance.
-
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Ethics & Legal Notice — Mandatory Acknowledgement

FOR EDUCATIONAL AND TRAINING USE ONLY. All network scanning, enumeration, and exploitation techniques documented in this report are to be practised exclusively within authorised, isolated laboratory environments. Performing any of the techniques described herein against systems, networks, or infrastructure without the **explicit prior written authorisation** of the system owner is a criminal offence in most jurisdictions worldwide.

Applicable Laws and Frameworks:

- **United States:** Computer Fraud and Abuse Act (CFAA), 18 U.S.C. § 1030 — unauthorised access to computer systems carries penalties of up to 10 years' imprisonment per count.
- **United Kingdom:** Computer Misuse Act 1990 — unauthorised access is an arrestable offence.
- **European Union:** Directive on Attacks Against Information Systems (2013/40/EU).
- **Pakistan:** Prevention of Electronic Crimes Act (PECA) 2016 — unauthorised access to information systems is a criminal offence under Sections 3 and 4.
- **International:** Budapest Convention on Cybercrime (ETS No. 185).